

Computer Vision – Lecture 12

CNNs V

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Course Outline

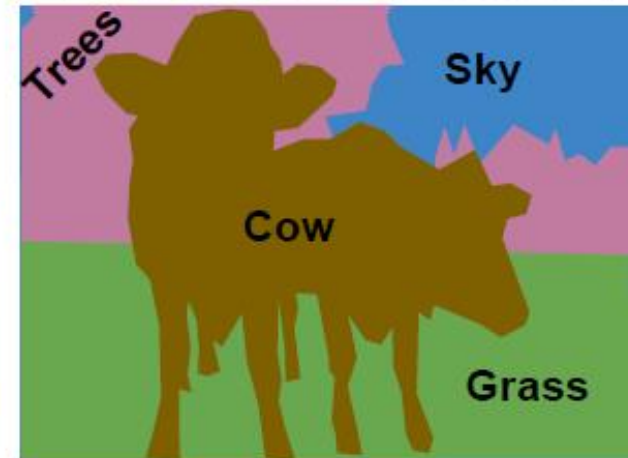
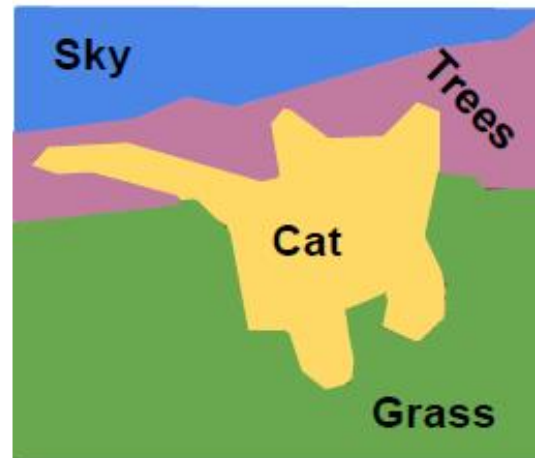
- Image Processing Basics
- Segmentation & Grouping
- Object Recognition & Categorization
 - Sliding Window based Object Detection
- Deep Learning for Vision
 - Convolutional Neural Networks (CNNs)
 - Deep Learning Background
 - CNNs for Object Detection
 - CNNs for Semantic Segmentation
 - CNNs for Matching
 - Transformers
- 3D Reconstruction

Topics of This Lecture

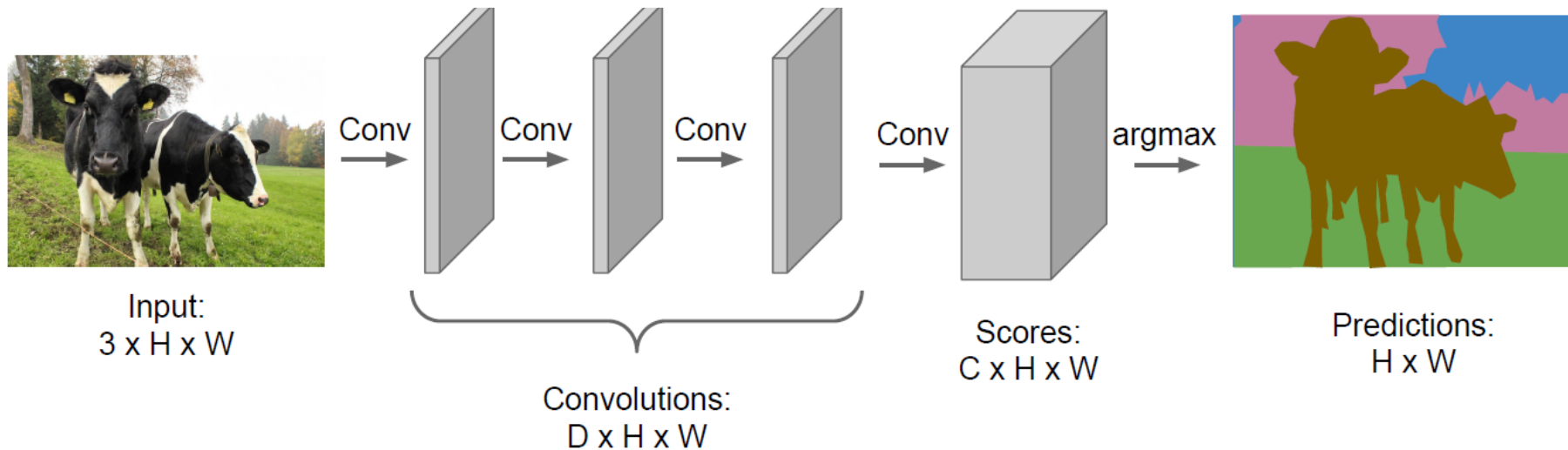
- CNNs for Segmentation
 - Recap: Encoder-Decoder architecture
 - Extensions
 - Segmentation architectures
- CNNs for Human Pose Estimation
 - 2D Body pose estimation
 - 3D Body pose estimation
- CNNs for Matching
 - Siamese networks
 - Triplet loss
 - Practical considerations
 - Applications

Recap: Semantic Segmentation

- Semantic Segmentation
 - Label each pixel in the image with a category label
 - Don't differentiate instances, only care about pixels
- Instance segmentation
 - Also give an instance label per pixel



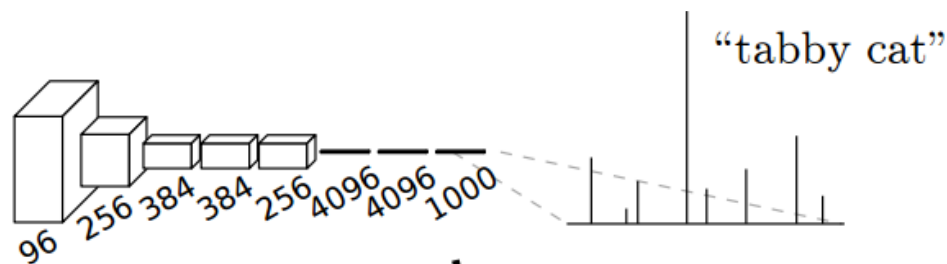
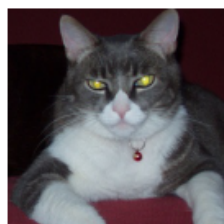
Recap: Fully-Convolutional Nets



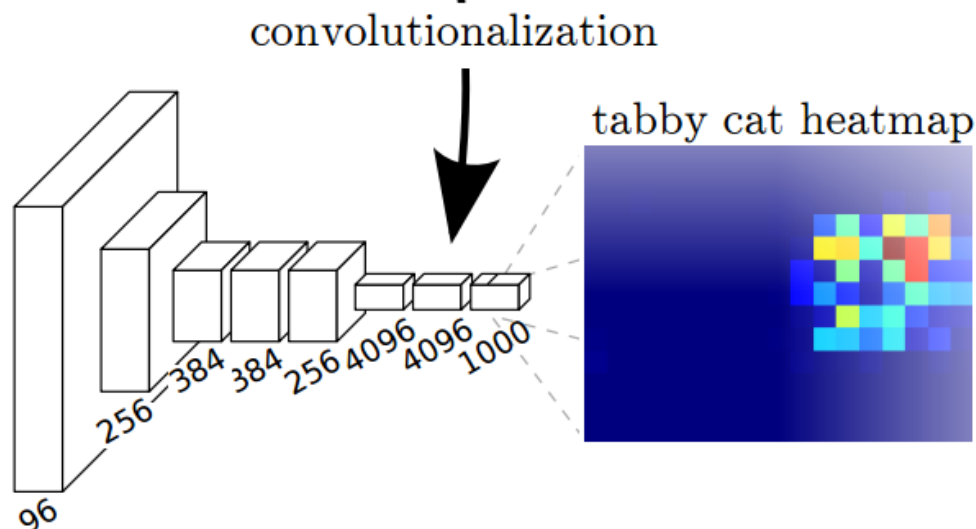
- Design a network as a sequence of convolutional layers
 - To make predictions for all pixels at once
 - **Fully Convolutional Networks (FCNs)**
 - All operations formulated as convolutions
 - Fully-connected layers become 1×1 convolutions
 - Advantage: can process arbitrarily sized images

Recap: CNNs vs. FCNs

- CNN



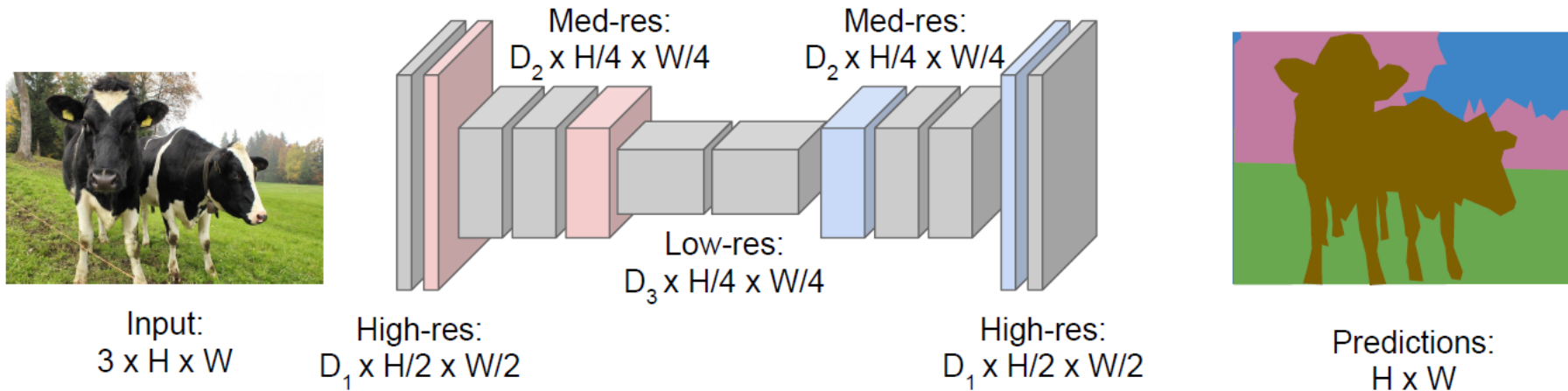
- FCN



- Intuition

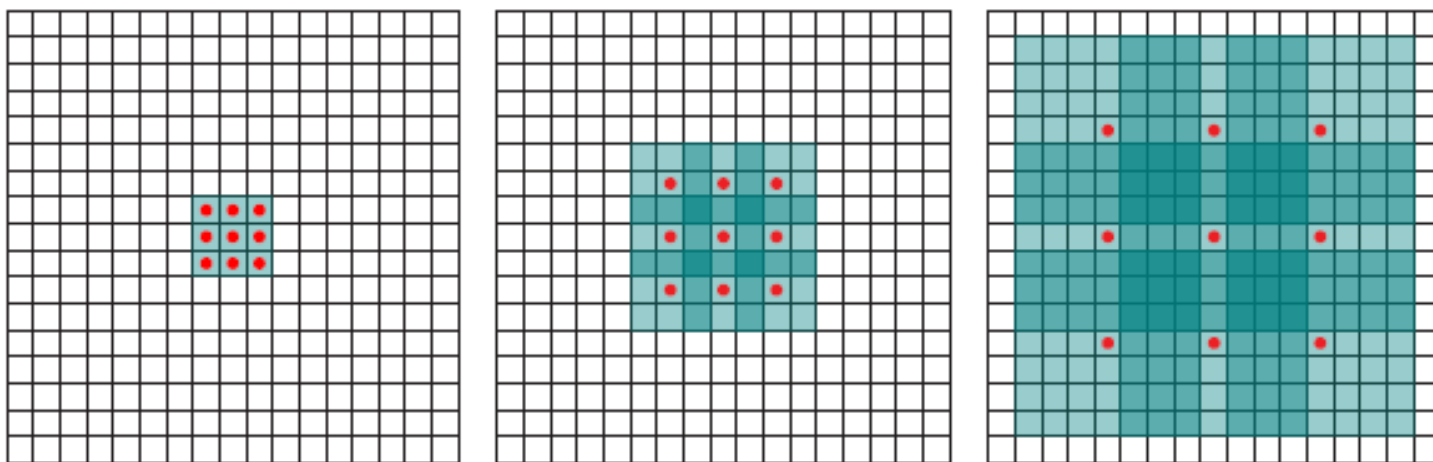
- Think of FCNs as performing a sliding-window classification, producing a heatmap of output scores for each class
- But: more efficient, since computations are reused between windows

Recap: Encoder-Decoder Architecture



- Design a network as a sequence of convolutional layers
 - With **downsampling** and **upsampling** inside the network!
 - **Downsampling**
 - Pooling, strided convolution
 - **Upsampling**
 - Unpooling or strided transpose convolution

Extension: Dilated Convolutions



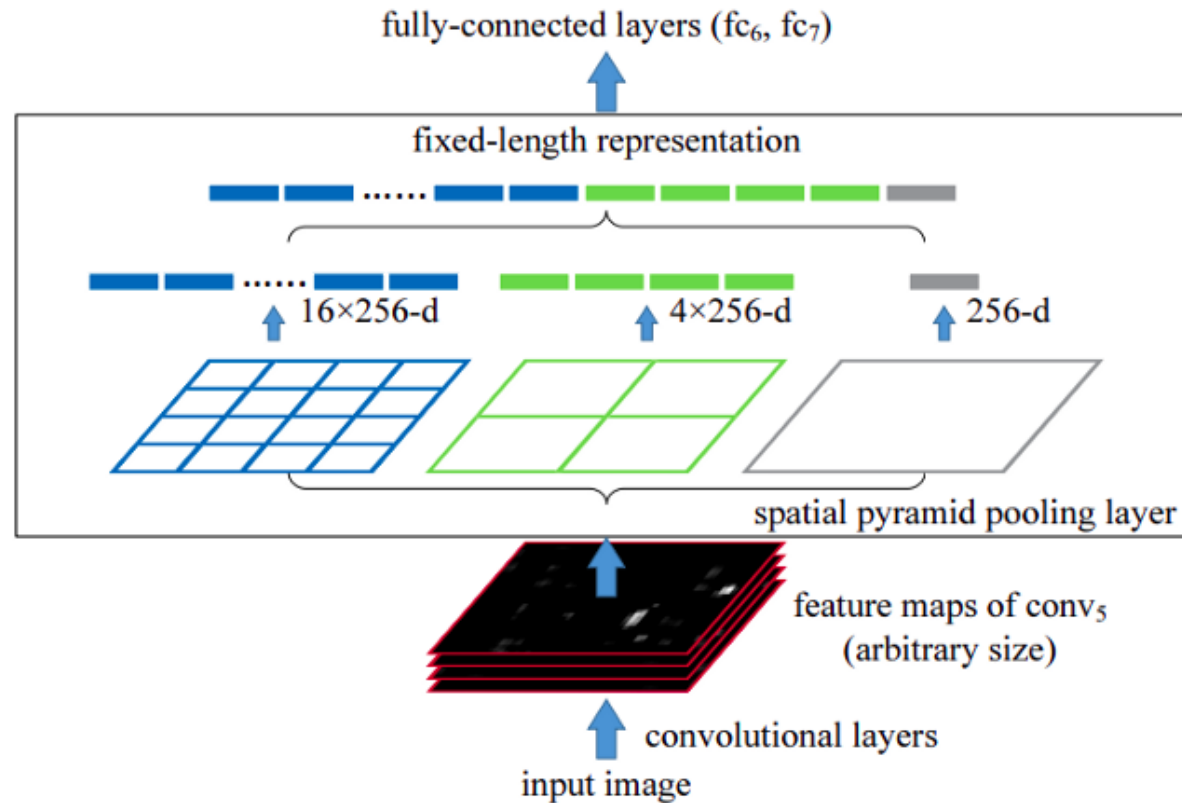
- Also known as: **à trous convolutions** or **atrous convolutions**
 - Simple way to increase the receptive field size without introducing additional parameters (filter size stays the same)

$$(F \star_l k)(\mathbf{p}) = \sum_{\mathbf{s} + l\mathbf{t} = \mathbf{p}} F(\mathbf{s}) k(\mathbf{t}) \quad \text{with dilation factor } l$$

- Technique plays a key role in the *algorithme à trous* for wavelet decomposition [Holschneider et al., 1987]

Fisher Yu, Vladlen Koltun, [Multi-Scale Context Aggregation by Dilated Convolutions](#), ICLR 2016

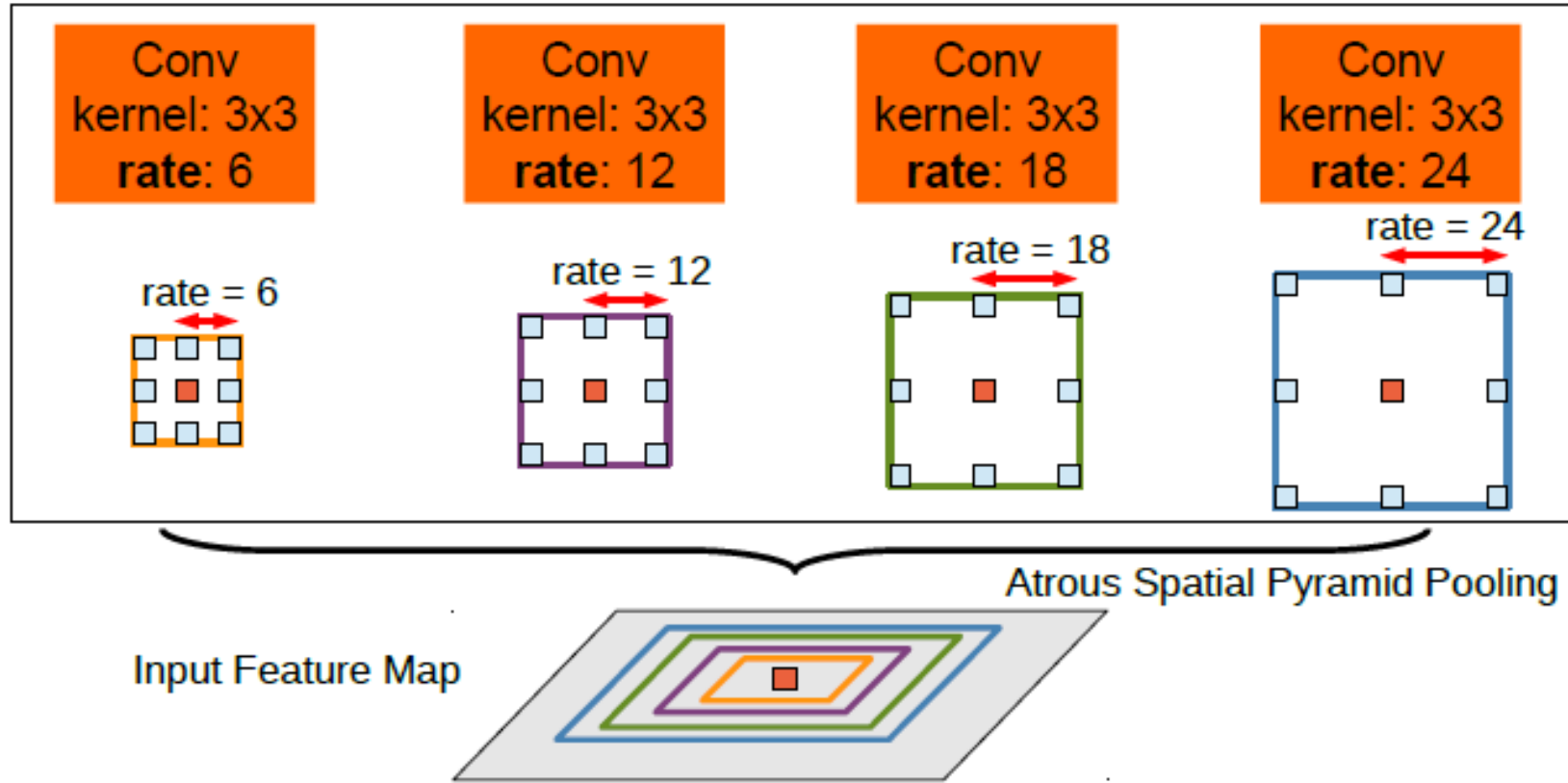
Extension: Spatial Pyramid Pooling (SPP)



- Multi-scale feature representation
 - Concatenated to a fixed-length vector

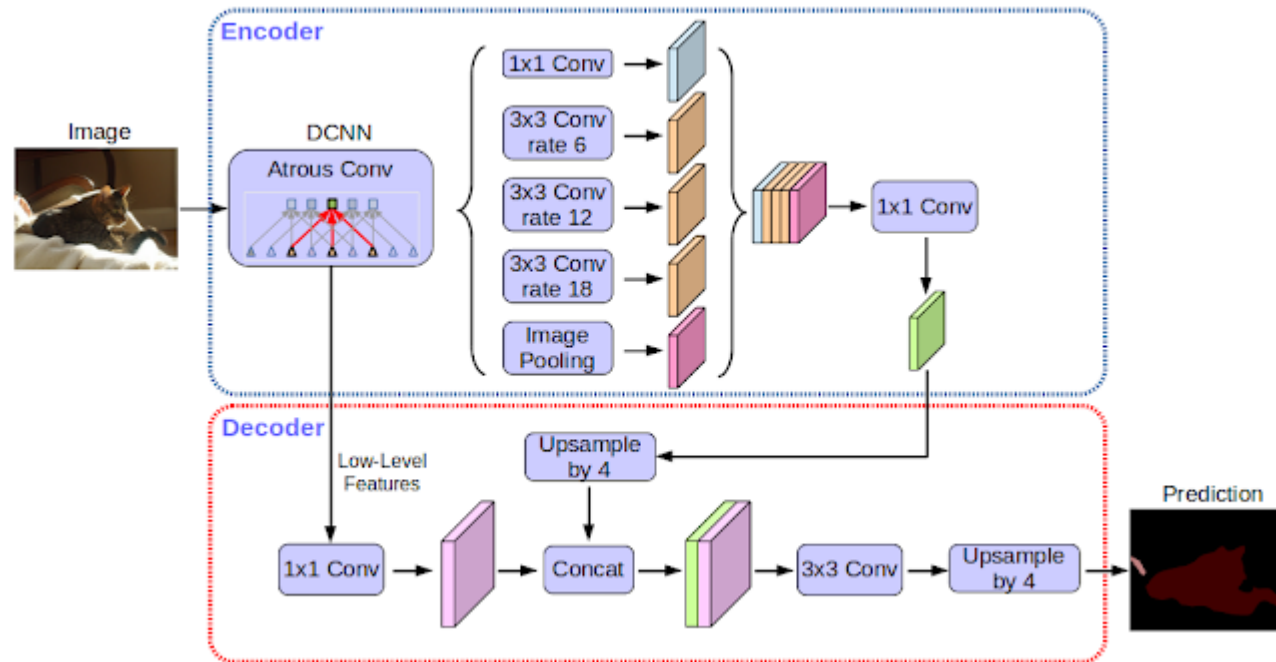
K. He, X. Zhang, S. Ren, J. Sun, [Spatial Pyramid Pooling in Deep Convolutional Networks for Visual Recognition](#), ECCV 2014.

Atrous Spatial Pyramid Pooling (ASPP)



- Extension of the SPP concept
 - Using **dilated convolutions** instead of regular convolutions
 - Effect: increased receptive field without extra computation

Example: DeepLabv3+



- Architecture designed for semantic segmentation
 - Uses **atrous spatial pyramid pooling (ASPP)** in the encoder
 - Simple decoder module
 - Depth-wise separable convolutions for efficiency

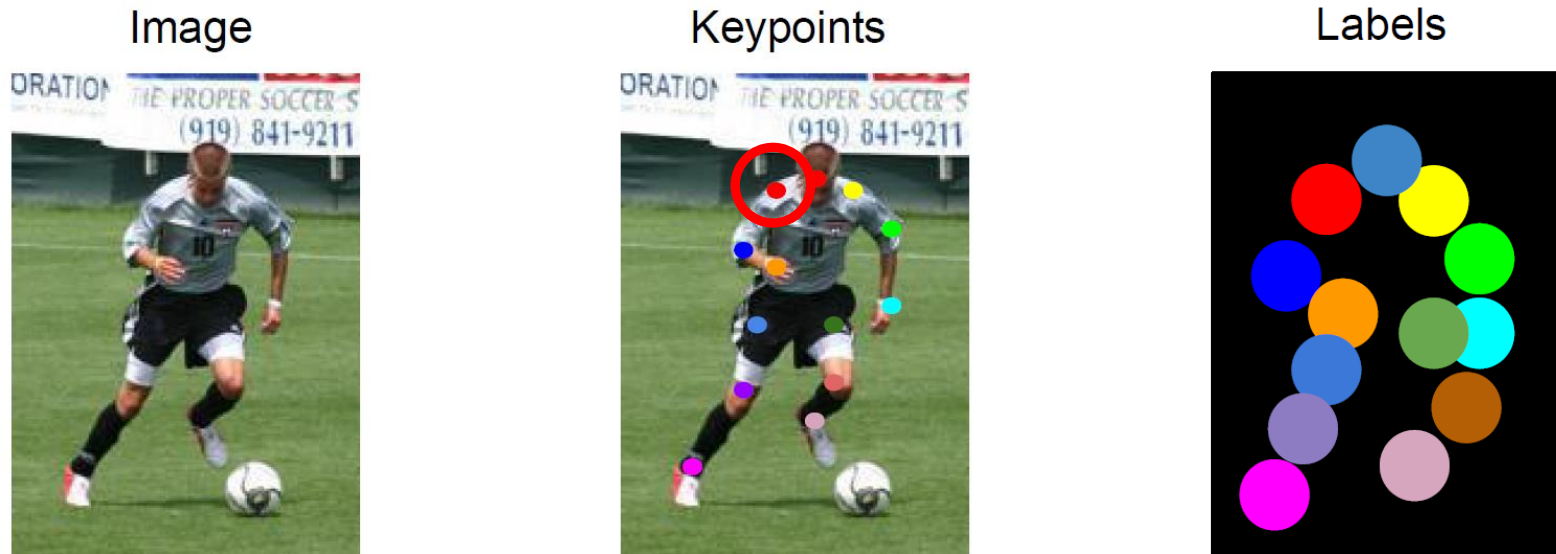
L-C. Chen, G. Papandreou, F. Schroff, H. Adam, [DeepLab: Semantic Image Segmentation with Deep Convolutional Nets, Atrous Convolutions, and Fully Connected CRFs](#), IEEE Trans. PAMI, 2017

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FCNs for Human Pose Estimation

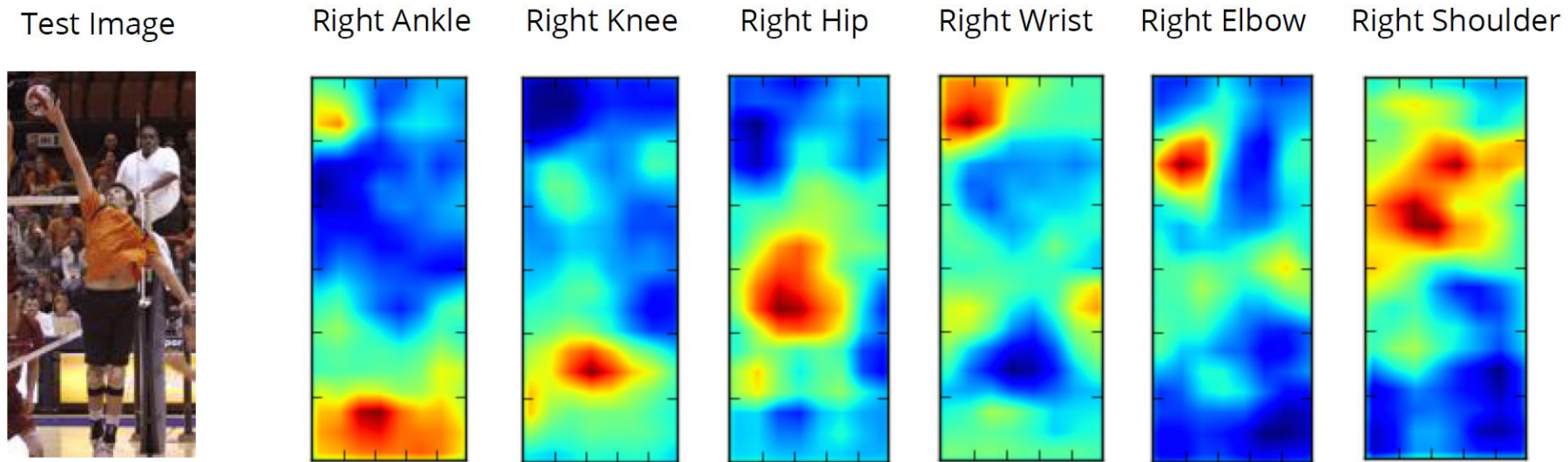
- Input data



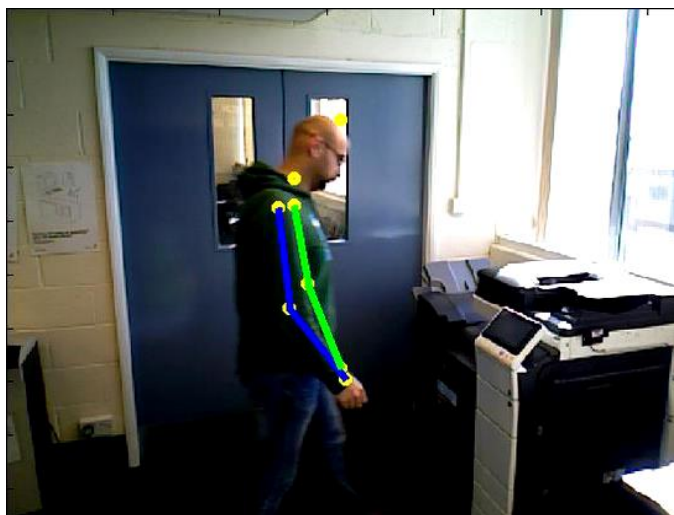
- Task setup

- Annotate images with keypoints for skeleton joints
- Define a target disk around each keypoint with radius r
- Set the ground-truth label to 1 within each such disk
- Infer heatmaps for the joints as in semantic segmentation

Heat Map Predictions from FCN



Example Results: Human Pose Estimation



Example Results: OpenPose

OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields

Zhe Cao, Gines Hidalgo, Tomas Simon, Shih-En Wei, Yaser Sheikh
Carnegie Mellon University

Z. Cao, G. Hidalgo Martinez, T. Simon, S.-E. Wei, [OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields](#), IEEE Trans. PAMI, 2019. ([code available here](#))

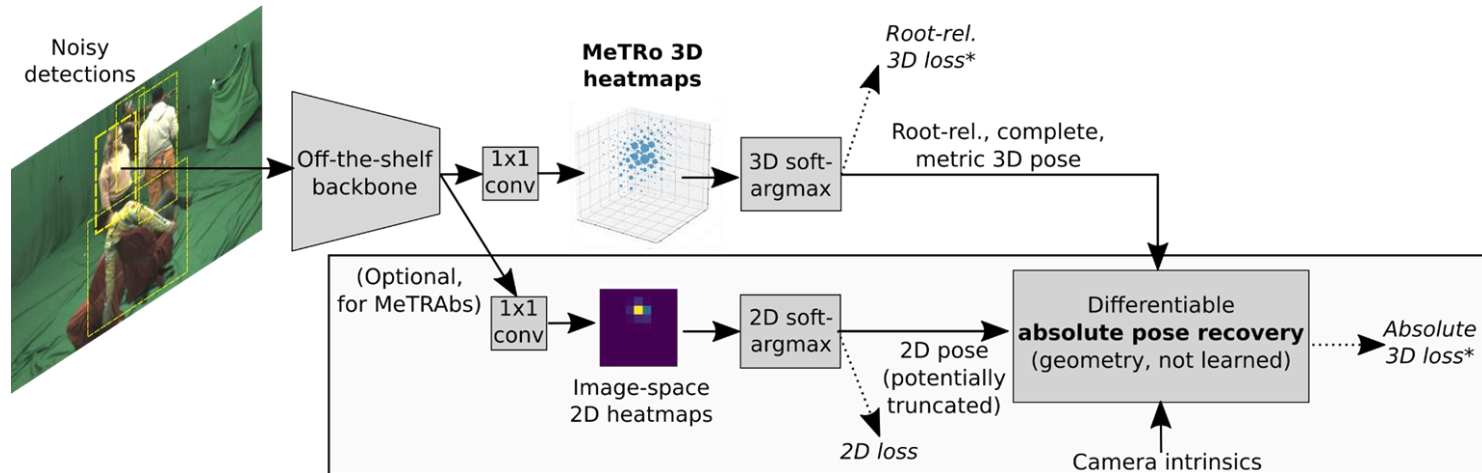
3D Human Body Pose Estimation



MeTRAbs

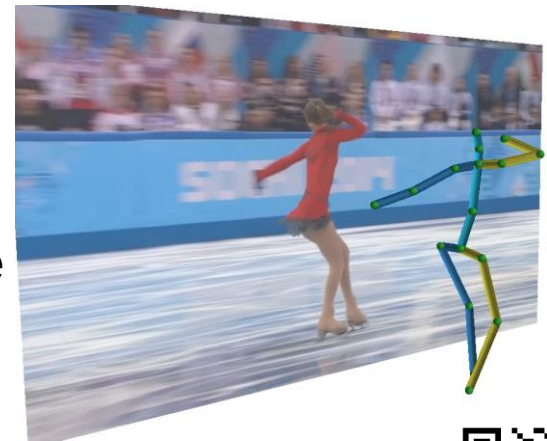


Architecture: CNN + Differentiable Geometry



- 3D Body Pose Estimation

- Metric-Scale Truncation-Robust Heatmaps for 3D human pose estimation (MeTRo)
- Robust to truncation, suitable for close range
- Very simple network design, extremely fast (can process >500 detections/s on a desktop GPU)



I. Sarandi, T. Linder, K.O. Arras, B. Leibe, [MeTRAbs: Metric-Scale Truncation-Robust Heatmaps for Absolute 3D Human Pose Estimation](#), IEEE Trans. BIOM, 2020.

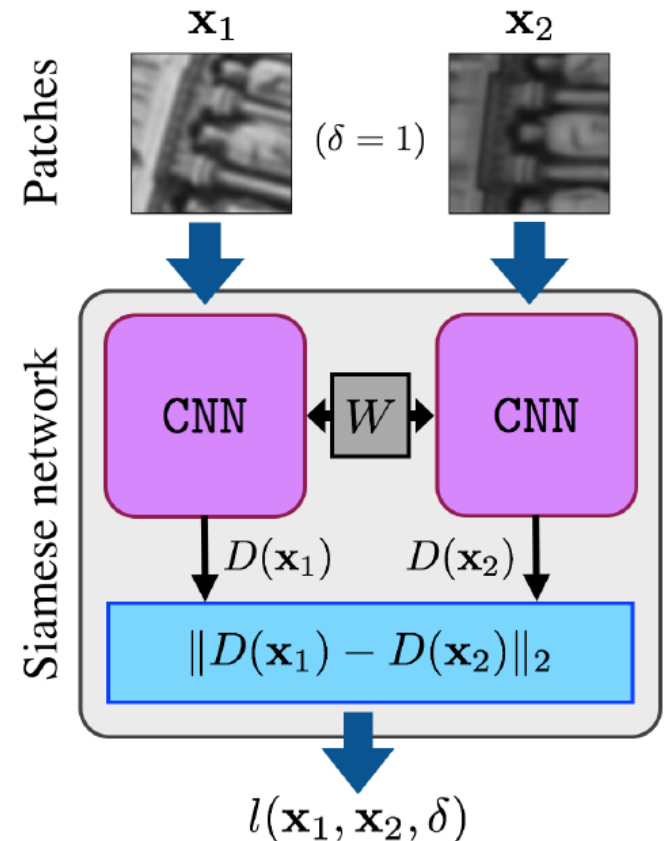


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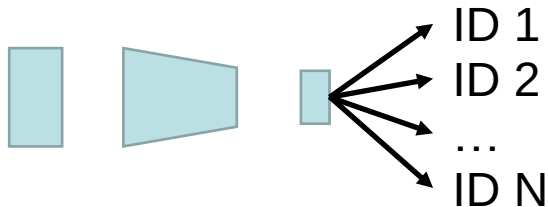
Learning Similarity Functions

- Siamese Network
 - Present the two stimuli to two identical copies of a network (with shared parameters)
 - Train them to output similar values if the inputs are (semantically) similar.
- Used for many matching tasks
 - Face identification
 - Stereo estimation
 - Optical flow
 - ...
- *How can we train a network to achieve this?*



Types of Models used for Matching Tasks

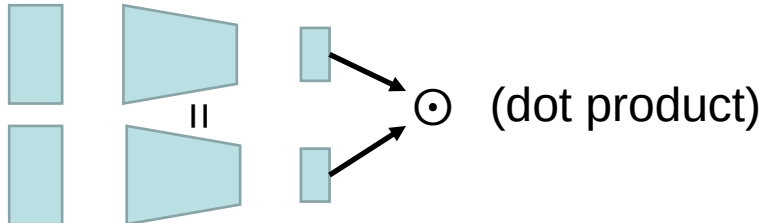
- Identification models (I)



Training

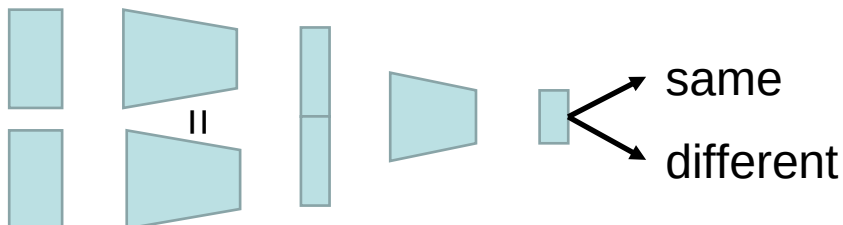
Multi-class
classification loss

- Embedding models (E)



Large-margin loss,
Triplet loss

- Verification models (V)



Two-class
classification loss

Triplet Loss Networks

- Learning a discriminative embedding
 - Present the network with triplets of examples

Negative



Anchor

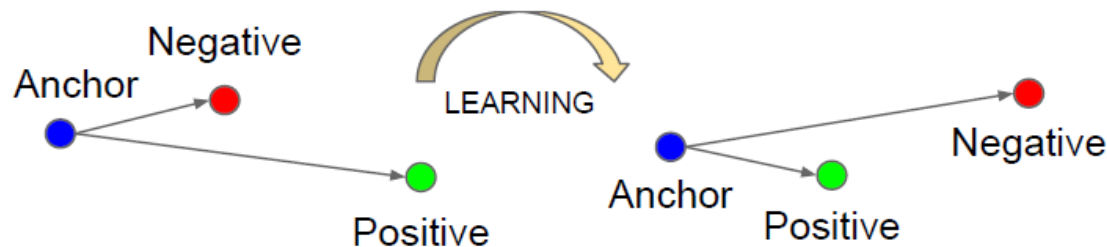


Positive



- Apply triplet loss to learn an embedding $f(\cdot)$ that groups the positive example closer to the anchor than the negative one.

$$\|f(x_i^a) - f(x_i^p)\|_2^2 < \|f(x_i^a) - f(x_i^n)\|_2^2$$



⇒ Used with great success in Google's FaceNet face identification

Triplet Loss – Practical Implementation

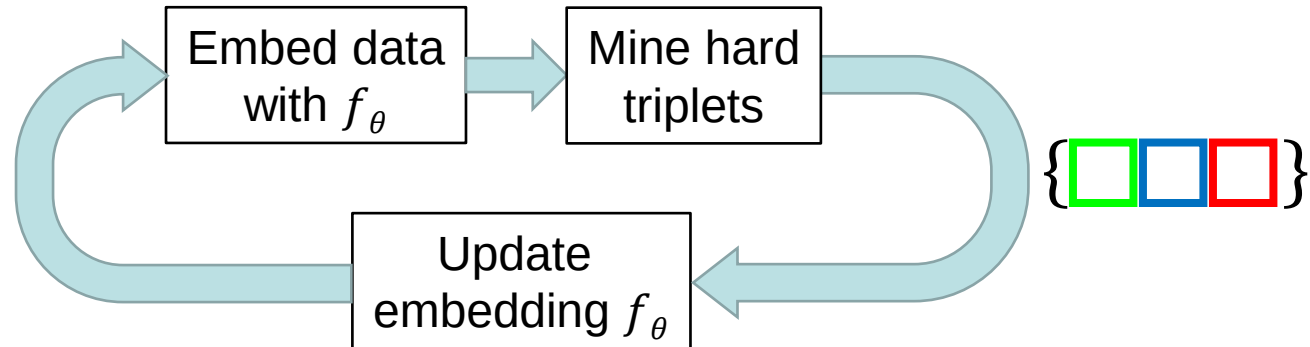
- Triplet loss formulation (using distances $D_{a,p} = d(x_a, x_p)$)

$$\mathcal{L}_{\text{tri}}(\theta) = \sum_{\substack{a,p,n \\ y_a=y_p \neq y_n}} [m + D_{a,p} - D_{a,n}]_+$$

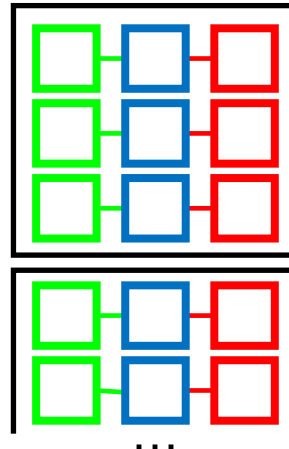
- Practical Issue: How to select the triplets?
 - The number of possible triplets grows cubically with the dataset size.
 - Most triplets are uninformative
 - ⇒ Mining hard triplets becomes crucial for learning.
 - ⇒ Actually want *medium-hard* triplets for best training efficiency
- Popular solution: Offline hard triplet mining
 - Process the dataset to find hard triplets
 - Use those for learning
 - Iterate

Offline Hard Triplet Mining

- Considerable effort needed



- Using the triplets for learning
 - Minibatch learning

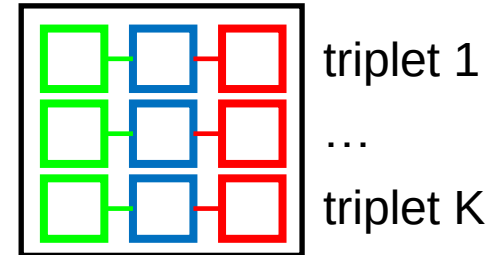


This is a very
wasteful design!

Better: Online Hard Triplet Mining

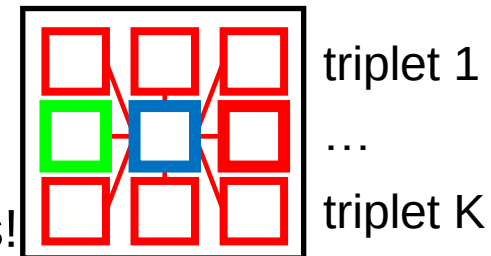
- Core idea

- The minibatch contains many more potential triplets than the ones that were mined!
- Why not make use of those also?



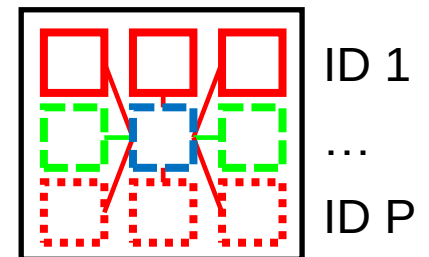
- Possible improvement

- Each member of another triplet becomes an additional negative candidate
- But: need both hard negatives *and* hard positives!



- Better design

- Sample K images from P classes (=people) for each minibatch
- Triplets are only constructed within the minibatch



Batch Triplet Loss Variants

- Batch All

[Ding et al., PR'15]

“Use all combinations of triplets within the batch”

$$\mathcal{L}_{\text{BA}}(\theta; X) = \sum_{i=1}^{\overbrace{P}^{\text{all anchors}}} \sum_{a=1}^{\overbrace{K}^{\text{all pos.}}} \sum_{\substack{p=1 \\ p \neq a}}^{\overbrace{K}^{\text{all negatives}}} \sum_{\substack{j=1 \\ j \neq i}}^{\overbrace{P}^{\text{all anchors}}} \sum_{n=1}^{\overbrace{K}^{\text{all pos.}}} \left[m - d_{j,a,n}^{i,a,p} \right]_+$$

$$d_{j,a,n}^{i,a,p} = D(f_{\theta}(x_a^i), f_{\theta}(x_p^i)) - D(f_{\theta}(x_a^i), f_{\theta}(x_n^j)).$$

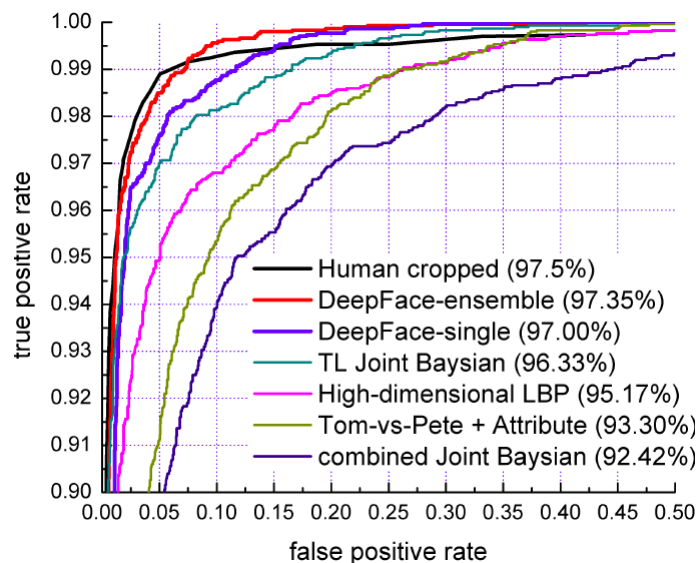
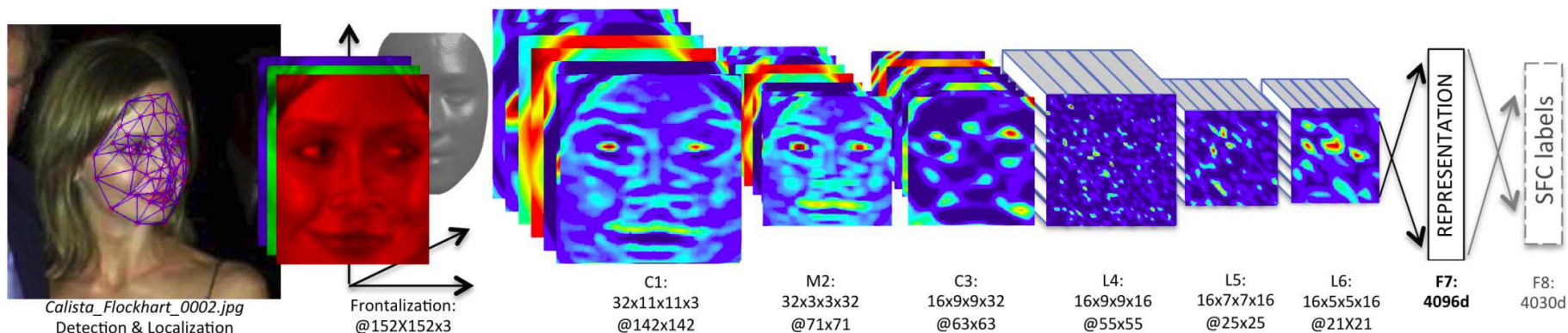
- Batch Hard

[Hermans et al., arXiv'17]

“For each sample, select the hardest positive and negative within the batch”

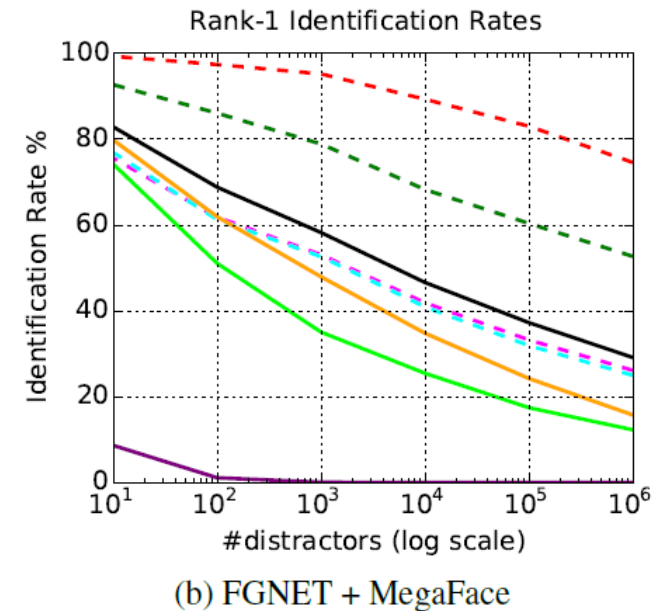
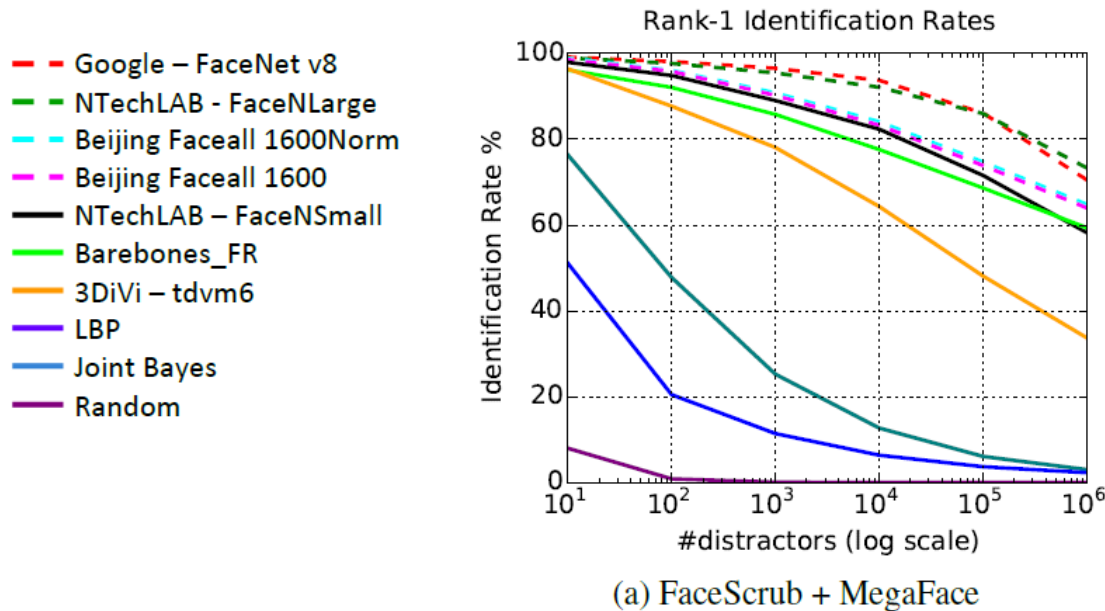
$$\mathcal{L}_{\text{BH}}(\theta; X) = \sum_{i=1}^{\overbrace{P}^{\text{all anchors}}} \sum_{a=1}^{\overbrace{K}^{\text{all pos.}}} \left[m + \underbrace{\max_{p=1 \dots K} D(f_{\theta}(x_a^i), f_{\theta}(x_p^i))}_{\text{hardest positive}} - \underbrace{\min_{\substack{j=1 \dots P \\ n=1 \dots K \\ j \neq a}} D(f_{\theta}(x_a^i), f_{\theta}(x_n^j))}_{\text{hardest negative}} \right]_+$$

Applications: Face Identification



Y. Taigman, M. Yang, M. Ranzato, L. Wolf, [DeepFace: Closing the Gap to Human-Level Performance in Face Verification](#), CVPR 2014

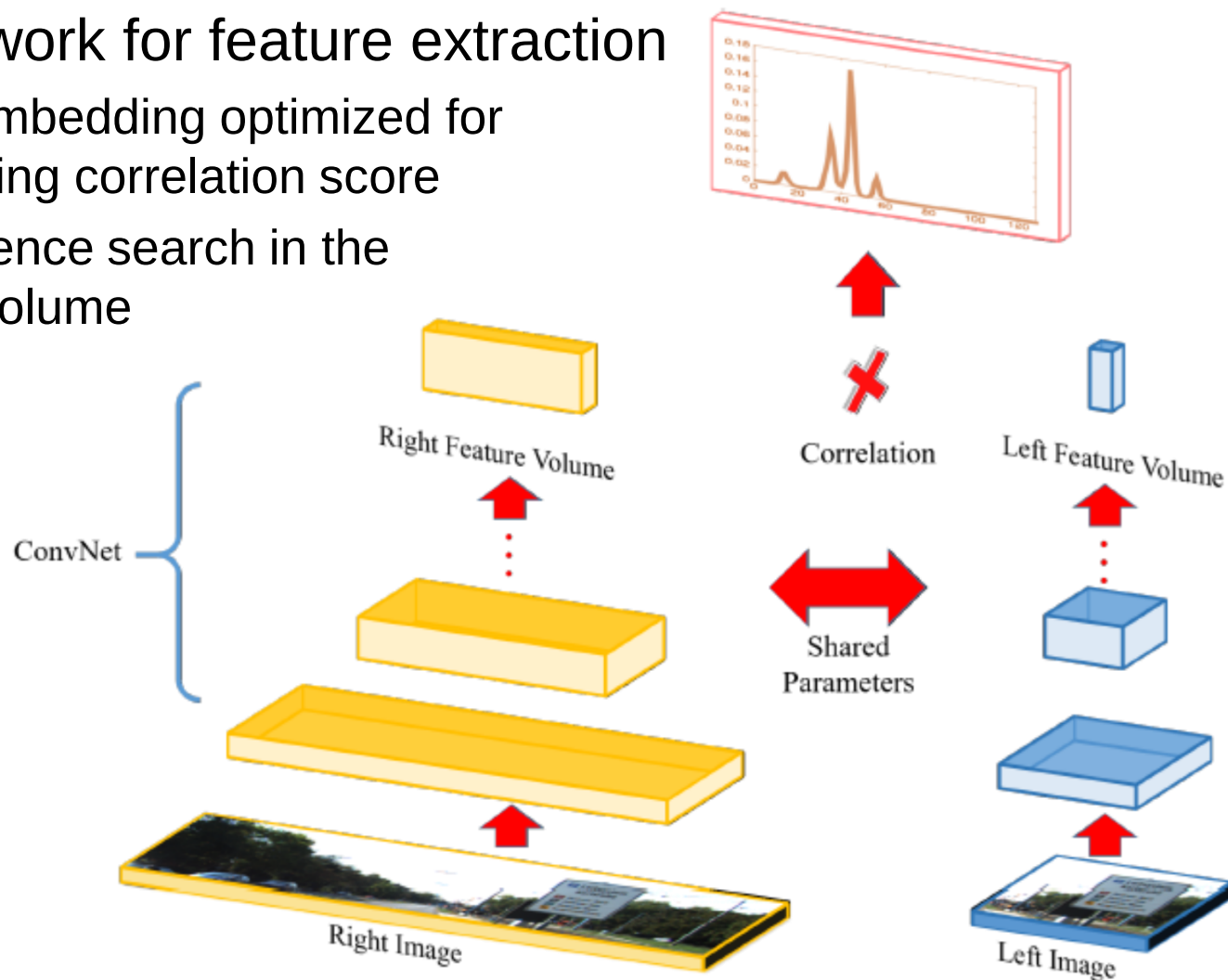
Face Identification Results



- Rank-1 identification rates of state-of-the-art algorithms
 - Performance with varying number of distractors in the gallery set
 - FaceNet: trained on 500M photos of 10M people

Applications: Stereo Matching

- Siamese network for feature extraction
 - Learns an embedding optimized for matching using correlation score
 - Correspondence search in the correlation volume



References: Computer Vision Tasks

- Semantic Segmentation

- J. Long, E. Shelhamer, T. Darrell, [Fully Convolutional Networks for Semantic Segmentation](#), CVPR 2015.
- O. Ronneberger, P. Fischer, T. Brox, [U-Net: Convolutional Networks for Biomedical Image Segmentation](#), MICCAI 2015
- V. Badrinarayanan, A. Kendall, R. Cipolla, [SegNet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation](#), arXiv 1511.00561, IEEE Trans. PAMI 2017.
- T-Y. Lin P. Dollar, R. Girshick, K. He, B. Hariharan, S. Belongie, [Feature Pyramid Networks for Object Detection](#), CVPR 2017.
- L-C. Chen, G. Papandreou, F. Schroff, H. Adam, [Rethinking Atrous Convolutions for Semantic Segmentation](#), arXiv 1706.05587 2017.

References: Computer Vision Tasks

- Human Body Pose Estimation

- A. Toshev, C. Szegedy, [DeepPose: Human Pose Estimation via Deep Neural Networks](#), CVPR 2014.
- S.E. Wei, V. Ramakrishna, T. Kanade, Y. Sheikh, [Convolutional Pose Machines](#), CVPR 2016.
- A. Newell, K. Yang, J. Deng, [Stacked Hourglass Networks for Human Pose Estimation](#), ECCV 2016.
- Z. Cao, T. Simon, S.-E. Wei, Y. Sheikh, [Realtime Multi-Person 2D Pose Estimation using Parts Affinity Fields](#), CVPR 2017.
- B. Xiao, H. Wu, Y. Wei, [Simple Baselines for Human Pose Estimation and Tracking](#), ECCV 2018. ([Code](#))